

Dominic Rousseau (he/him)

github.com/itsdombo

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(604) 618-3504

Skills

- **Programming Languages:** SystemVerilog, C/C++, Python, ARM Assembly, PLC Ladder Logic, Java, JavaScript
 - **Software Tools:** Git, Docker, PlatformIO, Android Studio, Bash, TimescaleDB, Figma, Quartus
 - **Libraries:** PyTorch, NumPy, Pandas, Matplotlib
 - **Hardware:** Raspberry Pi, Arduino, ESP32, Embedded Systems Development, Soldering, Hardware Debugging
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Education

University of British Columbia

September 2022 - Present

Bachelor of Applied Science, Computer Engineering

Work Experience

- **Software Engineering Intern - Backend/APIs** - SteepDeep, Vancouver January 2026 - March 2026
 - Designed a GraphQL-based API architecture to replace legacy REST endpoints using Firebase Data Connect
 - Built cloud-based backend services using Node.js and Google Cloud Platform for a SaaS application
 - Authored technical API documentation for third-party and frontend developer consumption in Github Projects
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Engineering Design Team Experience

- **Captain** - UBC Biological Internet of Things (BIoT) May 2024 - Present
 - Led 30 member team through full structural transition, ensuring continuous project development
 - Secured \$5,500 in funding through successful grant applications for applied fermentation R&D project
 - **Instrumentation Lead** - UBC Biological Internet of Things (BIoT) September 2022 - April 2024
 - Introduced GitHub workflows
 - Built modular, low-cost instrumentation tools to monitor pH, temperature, and dissolved oxygen content
 - Networked ESP32s and Arduinos for real-time sensor data collection and display **Grafana** and **TimescaleDB**
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Projects

- **Dynamic Autonomous Parking System (ROS2, PyTorch, YOLO, TensorRT, Jetson)** - Team Project
 - Built an autonomous parking system using ROS2, LiDAR, computer vision, and state-machine control
 - Trained a custom YOLO machine learning model on a self-labeled parking-sign dataset with transfer learning
 - Deployed real-time inference on NVIDIA Jetson Orin with TensorRT
 - **Virtual Fridge Android App (Kotlin/Java) - Grocery Tracker** - Team Project
 - Deployed a full-stack Android application to AWS
 - Configured cloud infrastructure and environments to support scalable backend services using Docker
 - Integrated Google Gemini API for produce recognition
 - Established CI/CD pipelines on GitHub with Codacy, achieving a barcode retrieval in under 3 seconds
 - Utilized Jira for burndown charts and agile sprints
 - **PLC Ladder Logic - Traffic Lights Finite State Machine** - Personal Project
 - Designed and implemented a finite state machine in Ladder Logic to simulate a real-world traffic light control system
 - Integrated timers and counters to enforce cycle durations and track the number of times the green light activated
 - **Self-Hosted Cloud & Game Server Infrastructure** – Personal + UBC ECESS
 - Enabled secure, remote access to 100GB+ of files across devices by running a Nextcloud instance on Ubuntu Linux
 - **Cluck Guard** - Personal Project
 - Engineered a battery-powered door for a chicken coop, programmed in C++ and powered by a servo and photoresistor
 - Published comprehensive open-source build guide on GitHub, enabling replication and reuse by hobbyists and makers
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Volunteer Experience

- **President** - Electrical and Computer Engineering Student Society May 2026 - Present
- **Vice President, Student Life** - Electrical and Computer Engineering Student Society May 2025 - April 2026
 - Organize orientations, game nights, and professional mixers for the department's undergraduate students
- **Server Infrastructure Manager** - Electrical and Computer Engineering Student Society November 2024 - Present
 - Self-host and maintain a Minecraft server for 20 people, replacing paid third-party hosting
 - Automated boot, backup, and crash recovery processes using Bash scripting and systemd on Ubuntu
 - Manage network configuration including port forwarding, firewall rules, and DNS to ensure 24/7 secure remote access
- **Creative Manager** - Engineering Undergraduate Society (UBC) September 2022 - April 2024
 - Composed patches, sports team logos, magazine covers, and posters for the Engineering Undergraduate Society drop sales for special edition items
- **Computer Electronics Repair Volunteer** - SPEC Repair Cafe February, 2023 - Present
 - Fix and service electronic devices for members of the public